SQL Joins



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Types of Joins

- Inner Join
  - Natural Join
- Left (Outer) Join
- Right (Outer) Join
- (Full) Outer Join
- Left (Outer) Join Excluding Inner Join
- Right (Outer) Join Excluding Inner Join
- (Full) Outer Join Excluding Inner Join
- Cross Join
- Equi-Join

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Sample Tables

TableA

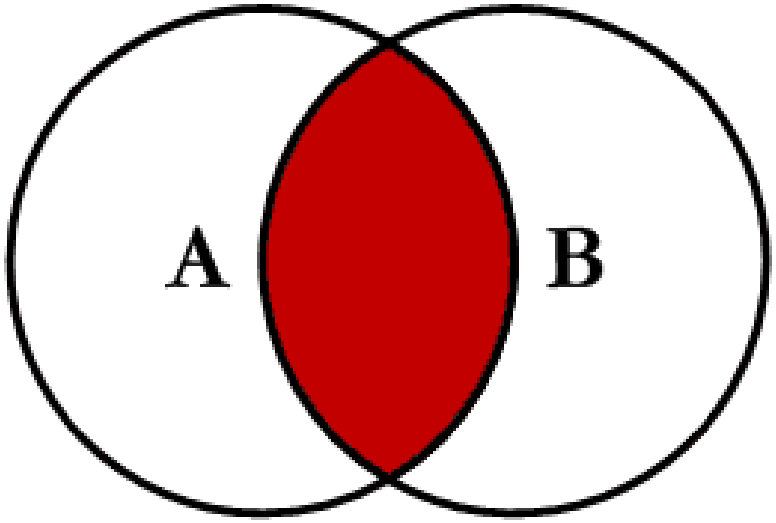
PK	Value
1	FOX
2	COP
3	TAXI
6	WASHINGTON
7	DELL
5	ARIZONA
4	LINCOLN
10	LUCENT

TableB

PK	Value
1	TROT
2	CAR
3	CAB
6	MONUMENT
7	PC
8	MICROSOFT
9	APPLE
11	SCOTCH

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Inner Join



- **Inner join** produces only the set of records that match in both Table A and Table B
- Most commonly used, best understood join

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Inner Join

TableA Value	PK	TableB PK	Value
FOX	1	1	TROT
COP	2	2	CAR
TAXI	3	3	CAB
WASHINGTON	6	6	MONUMENT
DELL	7	7	PC

SELECT * FROM TableA **INNER JOIN** TableB ON TableA.PK = TableB.PK

- This is the same as doing SELECT * FROM TableA, TableB **WHERE** TableA.PK = TableB.PK

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Inner Join (continued)

- Inner Joins do not have to use equality to join the fields
- Can use <, >, <>

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Inner Join (continued)

SELECT * FROM TableA **INNER JOIN** TableB ON TableA.PK > TableB.PK

TableA PK	Value	TableB PK	Value
2	COP	1	TROT
3	TAXI	1	TROT
3	TAXI	2	CAR
4	LINCOLN	1	TROT
4	LINCOLN	2	CAR
4	LINCOLN	3	CAB
5	ARIZONA	1	TROT
5	ARIZONA	2	CAR
5	ARIZONA	3	CAB
…	More…	Rows…	

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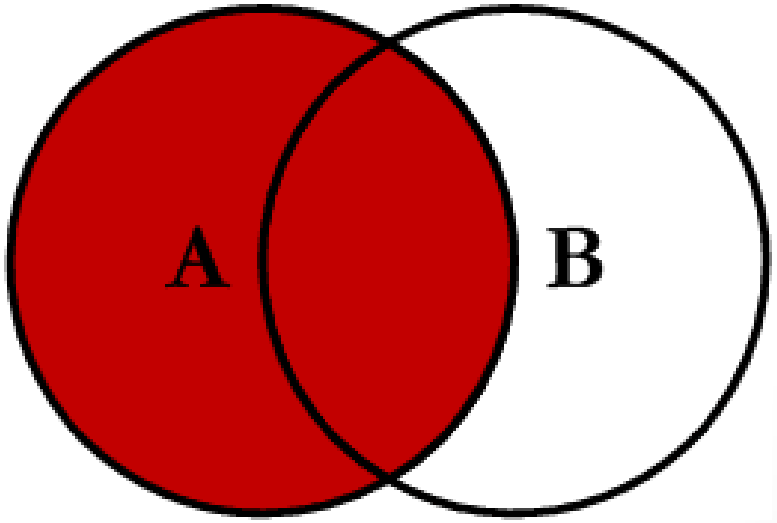
Inner Join/Natural Join

- A NATURAL join is just an inner equi-join where the join is implicitly created using **any** matching columns between the two tables
- Example:
  - SELECT * FROM TableA **NATURAL JOIN** TableB
  - Same results as inner equi-join?
  - Which columns match?

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Left Outer Join

- Left outer join produces a complete set of records from Table A, with the matching records (where available) in Table B. If there is no match, the right side will contain null.



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Left Outer Join

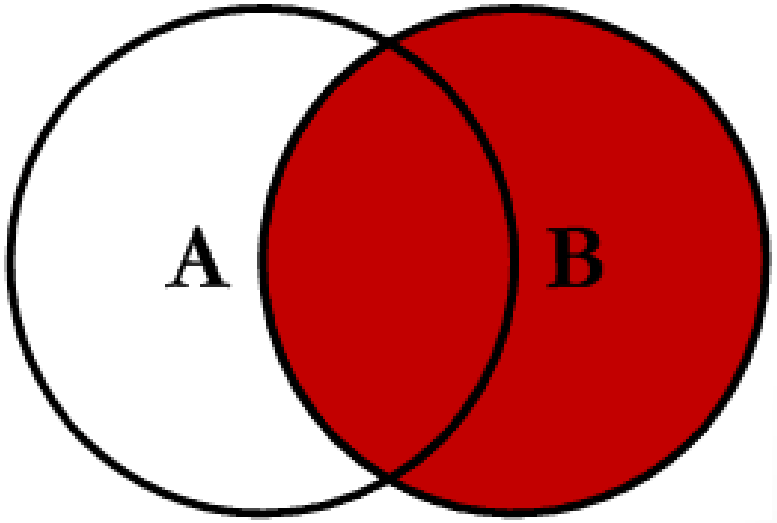
TableA Value	PK	TableB PK	Value
FOX	1	1	TROT
COP	2	2	CAR
TAXI	3	3	CAB
LINCOLN	4	NULL	NULL
ARIZONA	5	NULL	NULL
WASHINGTON	6	6	MONUMENT
DELL	7	7	PC
LUCENT	10	NULL	NULL

- SELECT * FROM TableA LEFT OUTER JOIN TableB ON TableA.PK = TableB.PK

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Right Outer Join

- Right outer join produces a complete set of records from Table B, with the matching records (where available) in Table A. If there is no match, the left side will contain null.



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Right Outer Join

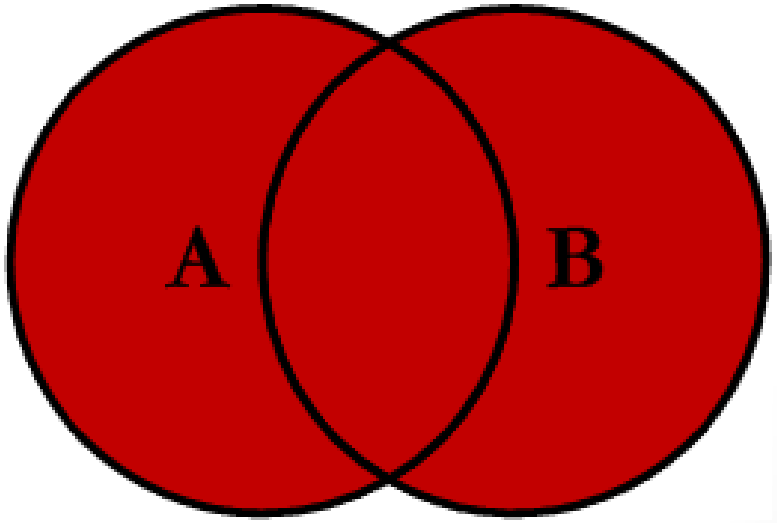
TableA Value	PK	TableB PK	Value
FOX	1	1	TROT
COP	2	2	CAR
TAXI	3	3	CAB
WASHINGTON	6	6	MONUMENT
DELL	7	7	PC
NULL	NULL	8	MICROSOFT
NULL	NULL	9	APPLE
NULL	NULL	11	SCOTCH

- SELECT * FROM TableA RIGHT OUTER JOIN TableB ON TableA.PK = TableB.PK

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Full Outer Join

- Full outer join produces the set of all records in Table A and Table B, with matching records from both sides where available. If there is no match, the missing side will contain null.



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Full Outer Join

TableA Value	PK	TableB PK	Value
FOX	1	1	TROT
COP	2	2	CAR
TAXI	3	3	CAB
LINCOLN	4	NULL	NULL
ARIZONA	5	NULL	NULL
WASHINGTON	6	6	MONUMENT
DELL	7	7	PC
LUCENT	10	NULL	NULL
NULL	NULL	8	MICROSOFT
NULL	NULL	9	APPLE
NULL	NULL	11	SCOTCH

- SELECT * FROM TableA FULL OUTER JOIN TableB ON TableA.PK = TableB.PK

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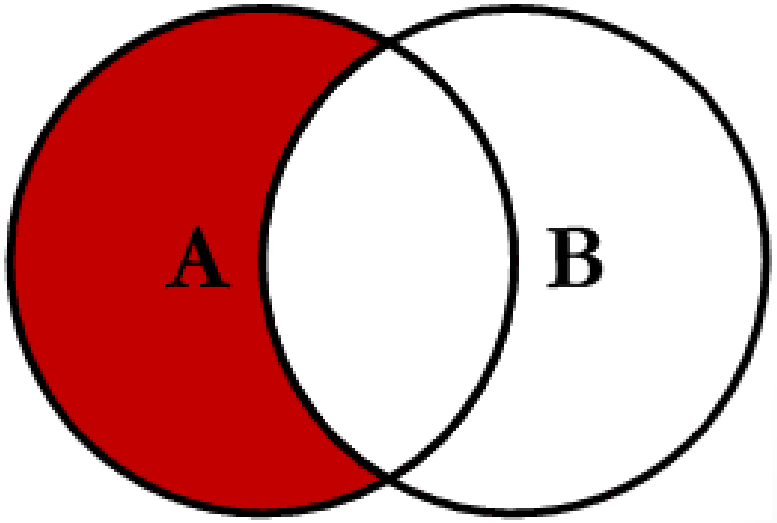
Full Outer Join in MySQL

- MySQL doesn't have FULL OUTER JOIN
- Simulate using UNION, LEFT and RIGHT JOINs
- SELECT * FROM TableA LEFT JOIN TableB
  ON TableA.PK = TableB.PK
  UNION
  SELECT * FROM TableA RIGHT JOIN TableB
  ON TableA.PK = TableB.PK

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Left Join Excluding Inner Join

- This query will return all of the records in the left table (table A) that do not match any records in the right table (table B).



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Left Join Excluding Inner Join

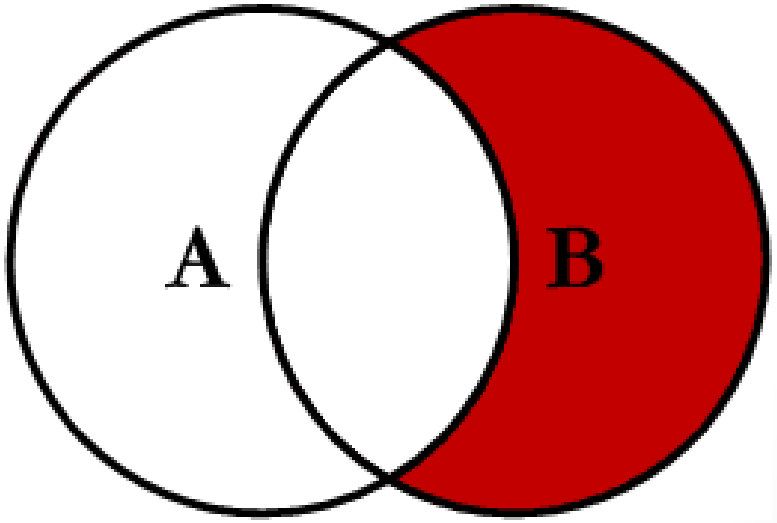
TableA Value	PK	TableB PK	Value
LINCOLN	4	NULL	NULL
ARIZONA	5	NULL	NULL
LUCENT	10	NULL	NULL

- SELECT * FROM TableA LEFT JOIN TableB
  ON TableA.PK = TableB.PK
  WHERE TableB.PK IS NULL
- Perform left outer join, then exclude the records we don't want from the right side via a where clause.

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Right Join Excluding Inner Join

- This query will return all of the records in the right table (table B) that do not match any records in the left table (table A).



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Right Join Excluding Inner Join

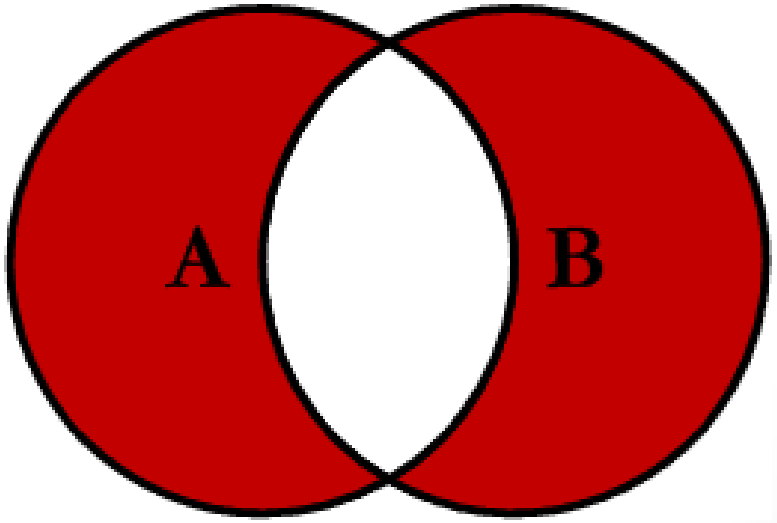
TableA Value	PK	TableB PK	Value
NULL	NULL	8	MICROSOFT
NULL	NULL	9	APPLE
NULL	NULL	11	SCOTCH

- SELECT * FROM TableA RIGHT JOIN TableB
  ON TableA.PK = TableB.PK
  WHERE TableA.PK IS NULL
- Perform right outer join, then exclude the records we don't want from the left side via a where clause.

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Full Outer Excluding Inner Join

- This query will return all of the records in Table A and Table B that do not have a matching record in the other table.
- (If you find a useful application, let me know! ☺)



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Full Outer Excluding Inner Join

TableA Value	PK	TableB PK	Value
NULL	NULL	8	MICROSOFT
NULL	NULL	9	APPLE
NULL	NULL	11	SCOTCH
LINCOLN	4	NULL	NULL
ARIZONA	5	NULL	NULL
LUCENT	10	NULL	NULL

- SELECT * FROM TableA FULL OUTER JOIN TableB ON TableA.PK = TableB.PK
  WHERE TableA.PK IS NULL OR TableB.PK IS NULL

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How Can We Do This in MySQL?

- MySQL doesn't have FULL OUTER JOIN
- Simulate using UNION, LEFT and RIGHT JOINs
- SELECT * FROM TableA LEFT JOIN TableB
  ON TableA.PK = TableB.PK
  WHERE TableB.PK IS NULL
  UNION
  SELECT * FROM TableA RIGHT JOIN TableB
  ON TableA.PK = TableB.PK
  WHERE TABLEA.PK IS NULL

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Cross Join

- A cross join is a Cartesian Product join – it is every record in Table A combined with every record in Table B.
- It gives the same results as not using a WHERE clause when querying two tables in MySQL
- SELECT * from TableA CROSS JOIN TableB
- SELECT * from TableA, TableB

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